Let’s study the problem and apply the design principles (DPs) you’ve learned, which are provided in appendix B. You’ll use DP 2 and DP 3 to guide you in defining the data structures. You’ll represent the design by using a contract diagram (DP 4) and the auction state transitions by using a finite state machine (FSM; DP 5), and you’ll use modifiers for any rules (DP 6) to be implemented by the smart contract. Applying the DPs provides a structured approach to starting blockchain solution development.

Do you see the pattern in the states of the auction? They’re similar to the states in the Ballot-Dapp. After all the DPs are applied, the contract diagram and the state transition diagram are as shown in figure 5.11. Before you start coding, make sure that you understand the process using the FSM on the left side of figure 5.11. In particular, note the `Bidding` and `Reveal` phases: blind bidding happens during the `Bidding` phase. A deposit is required for every bid; this deposit should be higher than the bid value. After all the blind bids are placed, the participants once again send their bid (open this time) during the `Reveal` phase. The winner is decided during the `Reveal` phase.

This problem is complex, but it exemplifies a pattern that is highly useful in many large systems, such as marketing and financial domains. Take your time to explore the ideas described here, and reuse them in your applications.